Task 14 — Map Product

Deterministic projection from Form Space to Army coordinates

Event	ITDX26 — U of Arizona Sierra Vista, 8–18 Sept 2026
Track	Algorithm-only, unclassified, laptop-based evaluation
Solicitation	W91RUSI-FCID260001 (SUB-007)
Submitter	Mycosoft, LLC · CAGE 9KR60 · UEI YK3ARVKJ77S9
PI / POC	Morgan Rockcoons, Founder & CEO · morgan@mycosoft.org · (619) 483-1077
Handling	UNCLASSIFIED // PROPRIETARY
Version	FormSpace overlay v1.0 — 2026-09-01

1. Overview — Map as Inspectable Rendering

The Map Product is a deterministic projection of the state estimate from Form Space onto Army-consumable geospatial coordinates, with uncertainty made visible. This task deliberately uses no machine learning in the critical path. That is a maturity signal: we know when not to use the model. The map renders what Tasks 12 and 13 produced; it does not reinterpret it.

2. Method — Deterministic Projection

Inputs from Tasks 12 and 13 are projected through a fixed function to produce:

- Device positions and health status.
- Direct and relayed communications paths (from DIRTNet routing).
- Hazard proxy with uncertainty envelope (95% CI shading).
- Tracks and standoff distances.
- Event timeline with adjudication labels.
- Evidence inspection panel (click-through to MINDEX records).
- Classification / handling banner (UNCLASSIFIED // PROPRIETARY for ITDX).

3. Fallback Ladder — Labeled in UI

The map explicitly labels its current data source in the operator UI at all times. The ladder is:

- medium — full model output.
- small — reduced-tier NLM output.
- quantized — INT8 quantized inference.
- distilled — student model output.
- classical — deterministic baseline only.
- prerecorded — signed replay from the field run.
- static — last-known-good snapshot with timestamp.

An Army SME can tell exactly which mode is active by reading the label. This removes the failure mode where a degraded system looks identical to a healthy one.

4. Outputs

- Rendered map (PNG + interactive HTML replay).
- Uncertainty overlay layer.
- Fallback-state label with timestamp.
- Merkle-verified evidence link for every rendered element.

5. Metrics

- Cold start ≤ 30 s (deterministic path is fastest).
- Refresh cadence ≤ 5 s per update.
- Uncertainty-envelope coverage vs empirical error on ground-truth stimulus.
- SME usability rating on a 5-point rubric.

6. Why No ML in the Critical Path

Learned components produce the underlying state estimate (Task 12) and the evidence graph (Task 13). The map itself is a rendering function. Keeping the rendering deterministic means an Army analyst can inspect exactly how a value on the map was computed by following the projection chain backward to MINDEX. That audit path is a governance property, not a performance limitation.

Morgan Rockcoons
Founder & CEO, Mycosoft, LLC
morgan@mycosoft.org · (619) 483-1077